The 2026 Capstone Project

AURA

3D Brain Tumor Segmentation

*Attention · **U**-Net · **Res**idual · **A**ll-modality*

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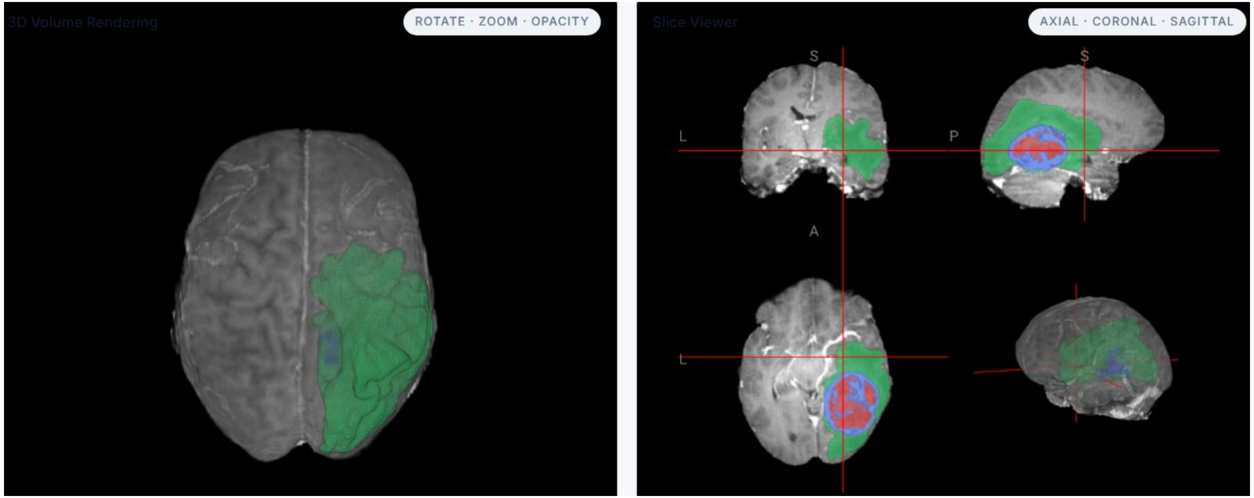
Outline

- Motivation
- Dataset
- Methodology
- Results
- Deployment
- Future Work
- Conclusion
- Q & A

01 Motivation

The Problem — and AURA

- Brain tumor are highly fatal and 5-year survival is ~ 30%
- Manual MRI segmentation takes up to 4 hours
- Lack of 3D visualization and instant volume metrics

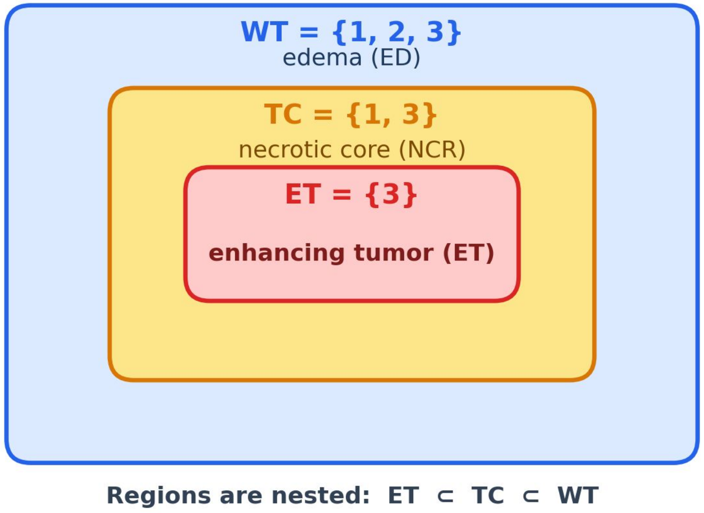


AURA: automated, 3D, instant volumetrics

02
Dataset

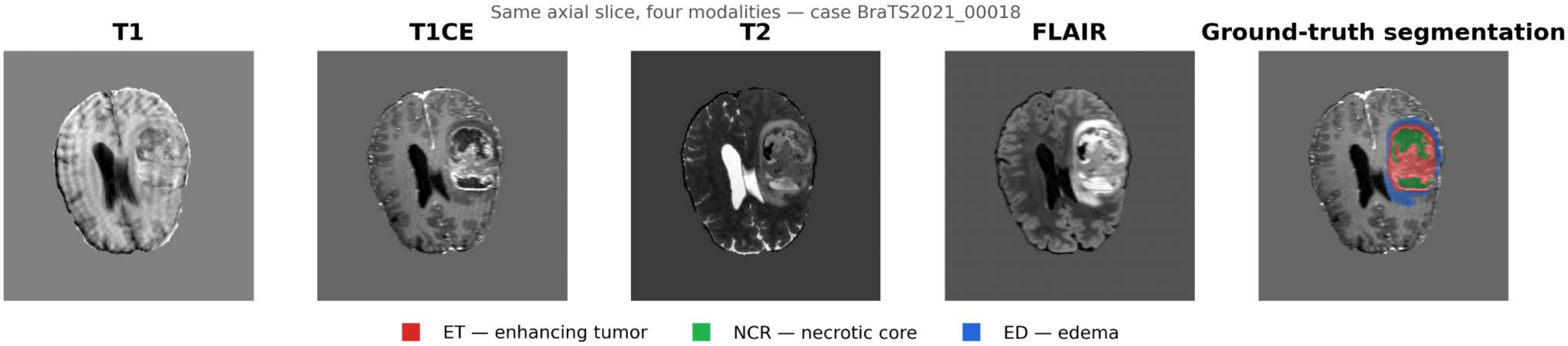
BraTS Overview

- Dataset: BraTS 2021
- Modalities: T1, T1CE, T2, FLAIR
- Classes: 0 BG, 1 NCR, 2 ED, 3 ET
- Train / Val: 1000 / 251 cases
- Shape: 240 × 240 × 155
- Input: 5-ch, 128^3 patches



MRI Modality

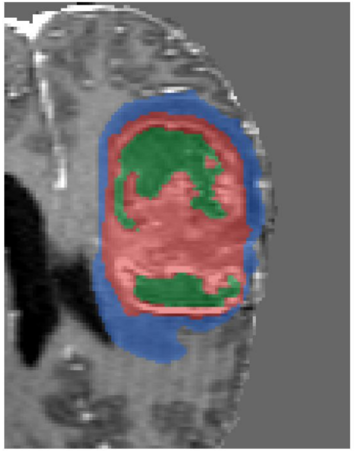
- No single modality shows the whole tumor
- T1CE highlights enhancing tumor
- FLAIR highlights edema
- T1 / T2 give anatomical structure



Clinical Region

- NCR = dead core · ET = active tumor · ED = infiltrative spread
- Resect ET and NCR to remove the tumor mass and relieve pressure; radiate ED for eliminate infiltrative cell

Real MRI scan — the three regions inside one tumor



NCR — necrotic core   ET — enhancing rim   ED — infiltrative edema

Tumor Core (TC) → Surgery

TC = NCR + ET *(solid tumor mass)*

Resect to debulk the tumor and relieve pressure

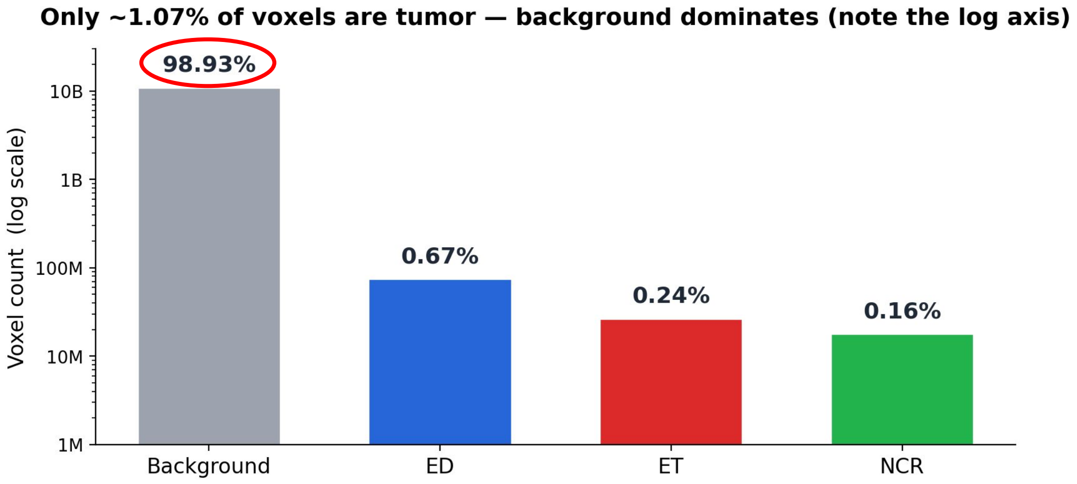
Whole Tumor (WT) → Radiotherapy

WT = NCR + ET + ED *(mass + infiltration)*

Irradiate the field to kill infiltrating cells

Class Imbalance

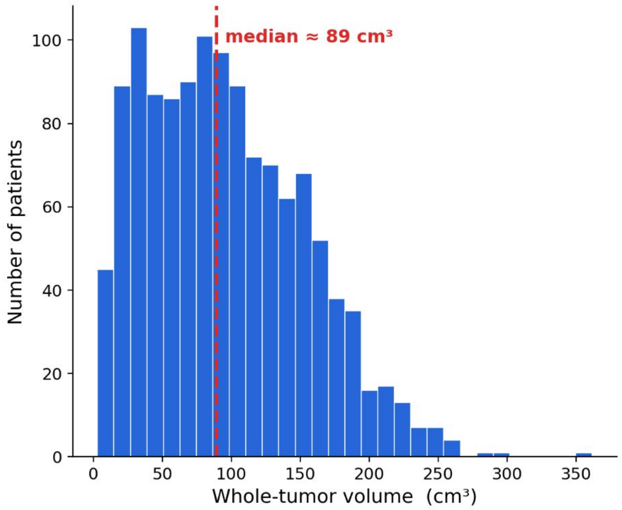
- Background : tumor ≈ 99 : 1; ET alone ≈ 412 : 1
- CE collapses to background, so we use region-wise Dice + Focal
- Same reason 50% of training patches are tumor-centered

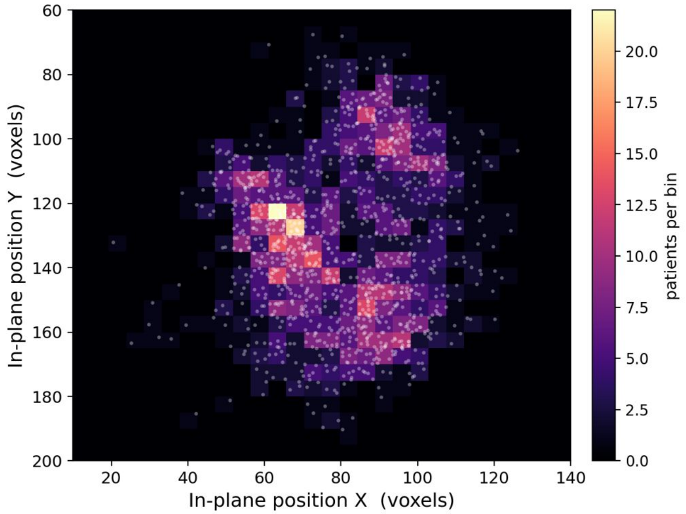


Background : ED ≈ 147 : 1 Background : ET ≈ 412 : 1 Background : NCR ≈ 617 : 1

Tumor Sizes & Location

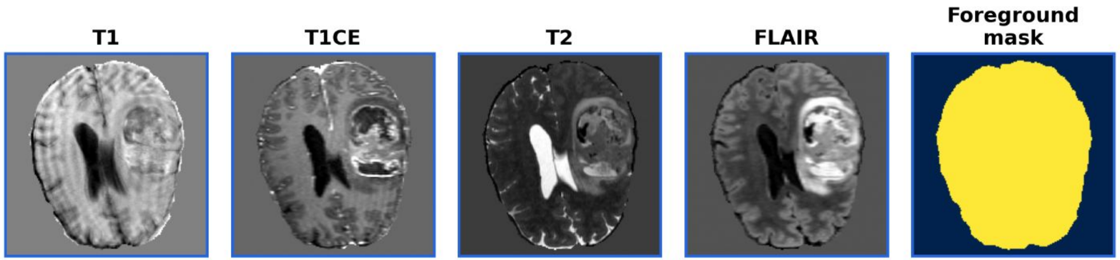
- Volume spans ~3–362 cm³ — a ~129× range (median ≈ 89 cm³)
- Centroids scattered brain-wide — no positional shortcut
- Leads to patch-based training + sliding-window inference

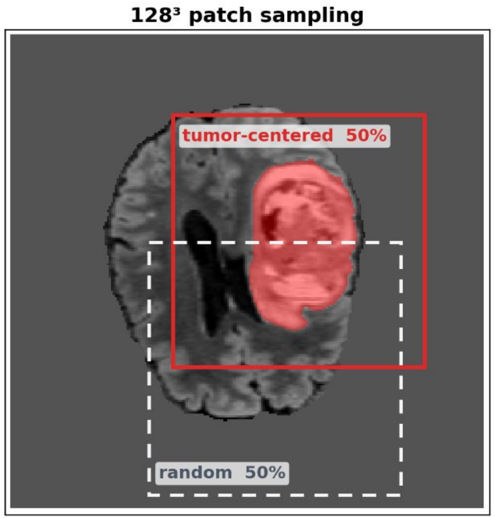




Model Input

- Input is 5-channel: T1 · T1CE · T2 · FLAIR + a foreground mask
- Z-score on brain voxels per modality, background zeroed
- Train on 128^3 patches — 50% tumor-centered, 50% random

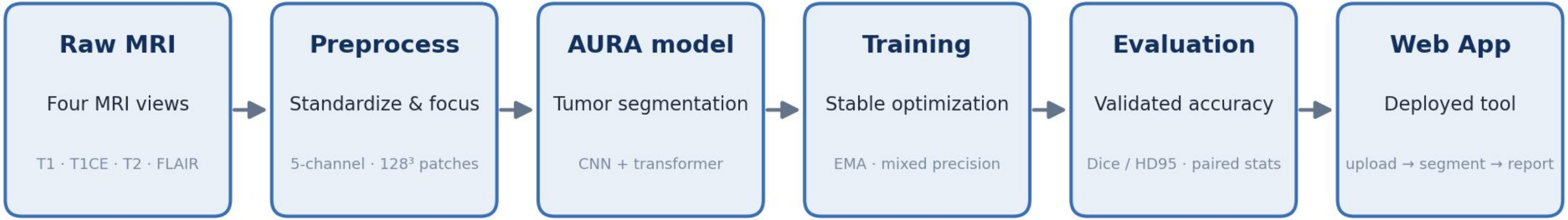




03
Methodology

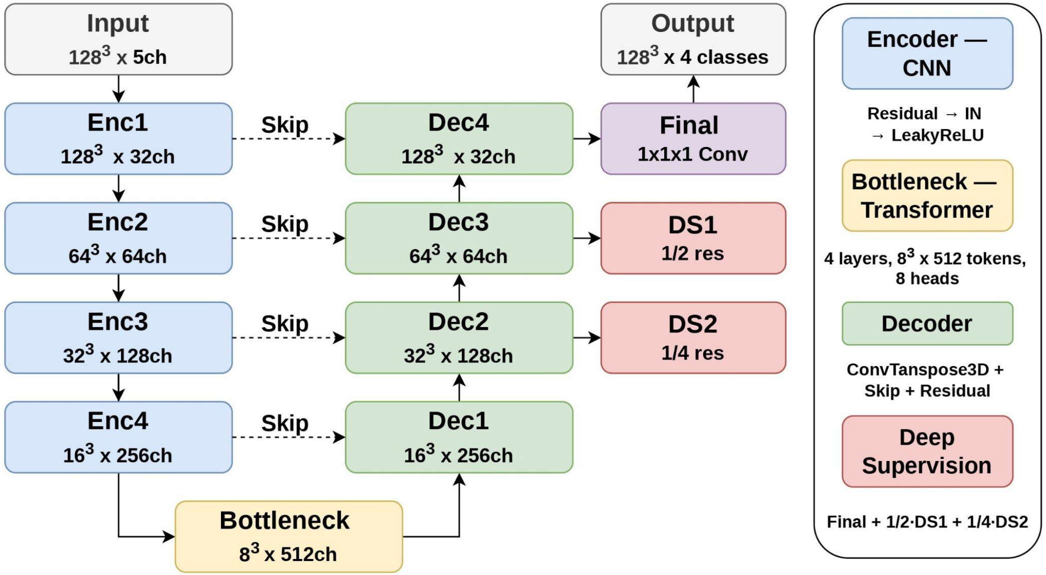
System Pipeline Overview

AURA is built as a complete end-to-end medical AI system, not just a model. Every stage from raw MRI to deployed web workstation is built, trained, and evaluated together.



AURA Architecture

- CNN + Transformer hybrid captures local texture and global 3D context
- Transformer at the 8^3 bottleneck only, efficient and fp16-stable
- 4-class softmax avoids region-overlap ambiguity of sigmoid outputs



Loss Function

- Region-wise Dice + Focal loss over ET, TC, WT
- Focal loss focuses training on hard, misclassified voxels
- Deep supervision propagating gradients through decoder stages

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{seg}}(\hat{y})}_{\text{full res}} + \underbrace{\tfrac{1}{2}\,\mathcal{L}_{\text{seg}}(\hat{y}_{DS_1})}_{1/2 \text{ res}} + \underbrace{\tfrac{1}{4}\,\mathcal{L}_{\text{seg}}(\hat{y}_{DS_2})}_{1/4 \text{ res}}$$

$$\mathcal{L}_{\text{seg}} = \sum_{r \in \{\text{WT}, \text{TC}, \text{ET}\}} (\text{Dice}_r + \text{Focal}_r)$$

$$\text{Dice}_r = 1 - \frac{2\sum p_r g_r}{\sum p_r + \sum g_r} \qquad \text{Focal}_r = -\sum (1 - p_t)^{\gamma} \log p_t, \quad \gamma = 2$$

Training Recipe

Stable optimization for 3D medical image segmentation.

Component	Setting
Optimizer	Adam, lr 1e-4
Effective batch size	32 (batch 2 × accum 16)
Epochs	300
AMP	fp16
EMA	decay 0.999
Gradient clipping	max norm 1.0
Patch sampling	50% tumor-centered

04
Results

Comparison Models

Baseline

Modules:

3D Residual U-Net

Role:

Lower bound

Complex

Modules:

3D Residual U-Net

+ Modality

+ Cross modal attention

+ Spectral Swin

+ Frequency filter

+ Uncertainty head

+ Boundary head

+ Multi-scale fusion

Role:

Upper bound

AURA

Modules:

3D Residual U-Net

+ Transformer bottleneck

Role:

Hero model

Evaluation Protocol

- AURA is evaluated across accuracy, calibration, and uncertainty
- TTA — 8-way flip, axes (D, H, W)
- MC-Dropout — 20 forward passes with Dropout active

Category	Metric	Measures
Segmentation	Dice	Overlap between prediction and ground truth
Segmentation	HD95	Worst-case boundary error (mm)
Calibration	ECE	How well confidence matches accuracy on tumor voxels
Uncertainty	AURC	Whether the model knows when it is wrong

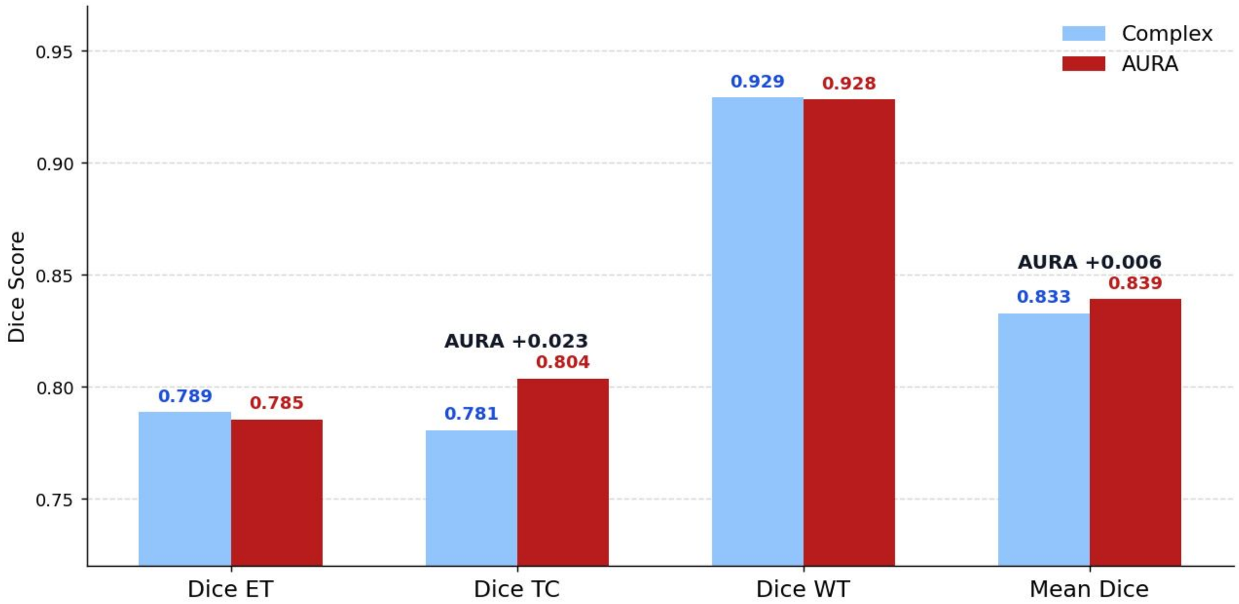
Evaluation Result — AURA Wins

- AURA achieves the highest Mean Dice (0.839), outperforming both Baseline and Complex
- Best calibration and uncertainty – lowest ECE (0.100) and AURC (0.175) across all models

Model	Mean Dice	Dice ET	Dice TC	Dice WT	HD95 ↓	ECE ↓	AURC ↓
Baseline	0.825	0.765	0.788	0.923	6.750	0.106	0.179
Complex	0.833	**0.789**	0.781	**0.929**	**5.95**	0.110	0.191
AURA	**0.839**	0.785	**0.804**	**0.928**	**5.96**	**0.100**	**0.175**

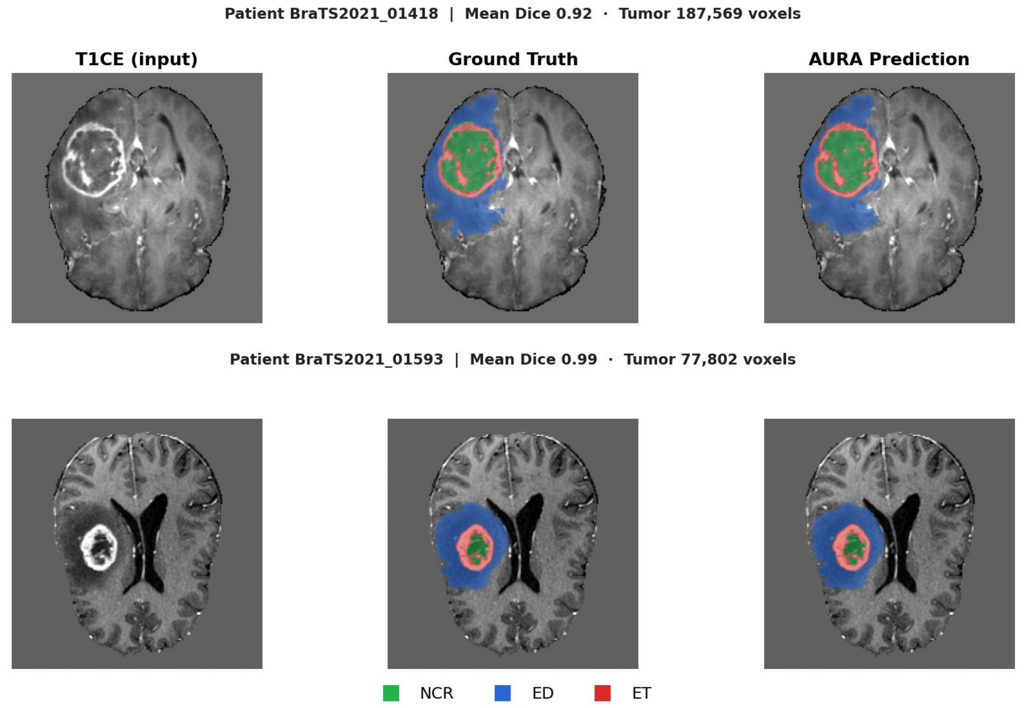
Less is More

- More modules ≠ better performance, Complex has 8, AURA has 2
- AURA wins TC by +0.023 — the most clinically critical region



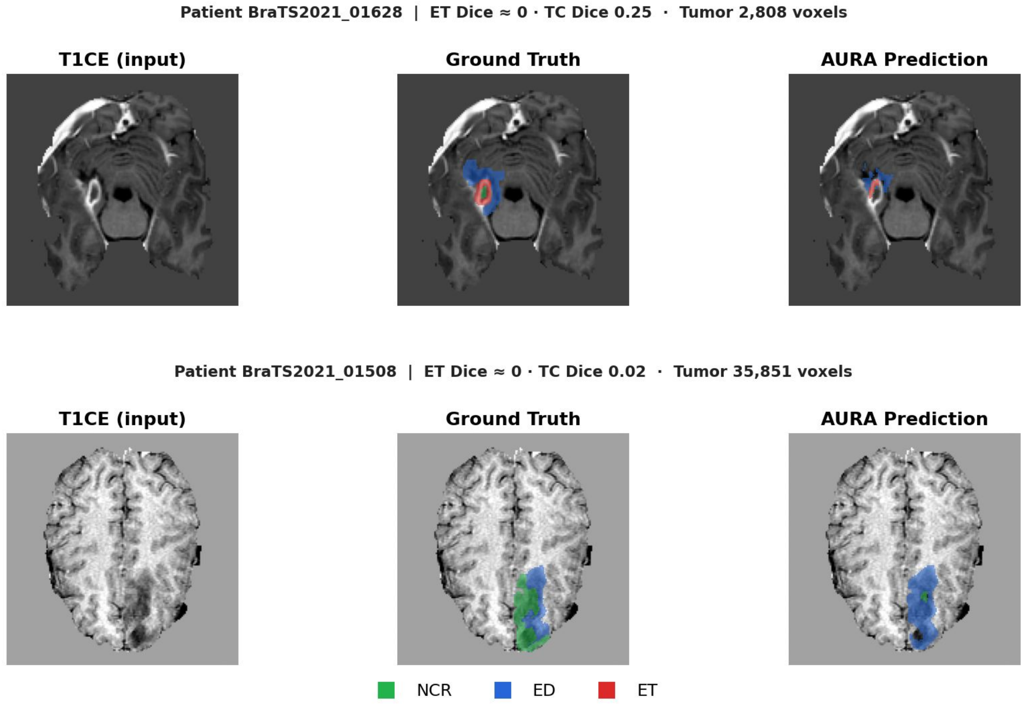
Qualitative — Success

AURA correctly identifies all three tumor regions across cases.



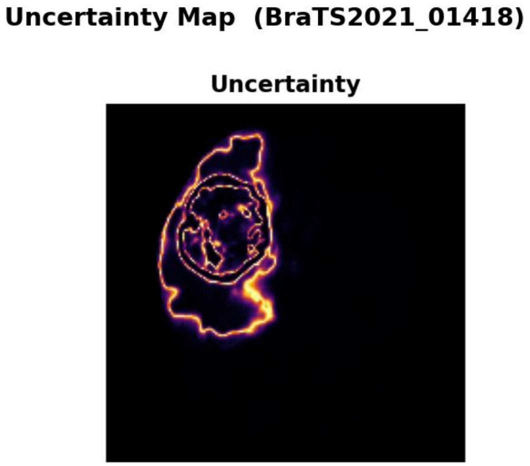
Qualitative — Failure

AURA struggles when ET is small and NCR is misread as edema.

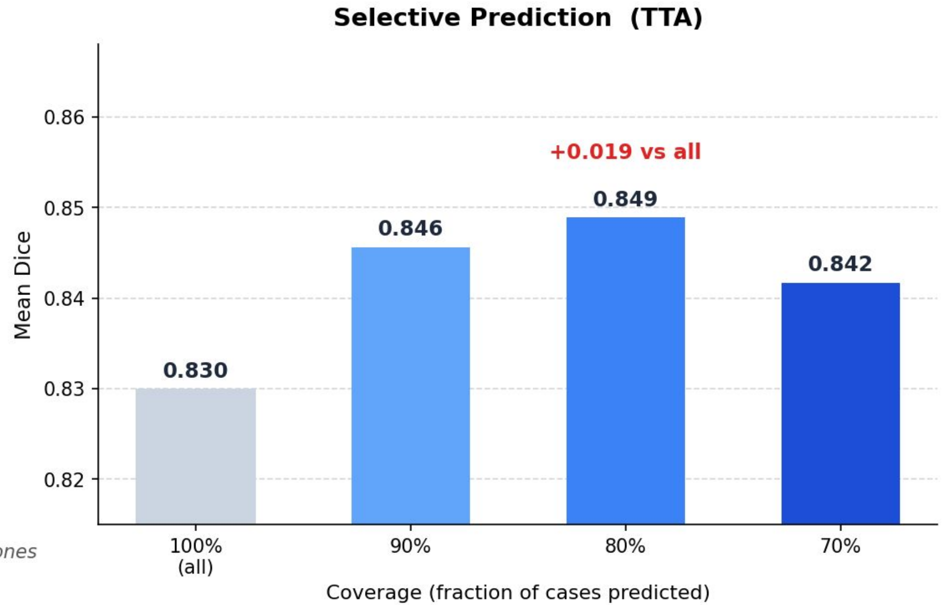


Trustworthiness

- AURA assigns lower confidence to cases it is likely to get wrong
- Skipping the least confident 20% raises Dice from 0.830 to 0.849



High uncertainty concentrates at region boundaries and error zones



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Feng-Yu Hsu

Complexity

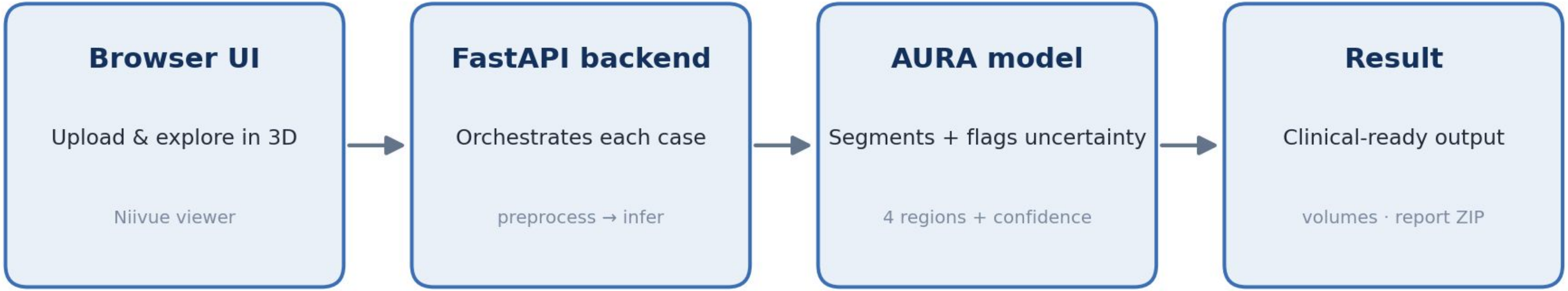
At 45M parameters and 242 ms per scan, AURA fits on a single consumer GPU with no compromise on accuracy.

Metric	AURA
Parameters	45 M
FLOPs	612 GFLOPs
GPU Memory	1.9 GB
Model Size	172 MB
Inference Time	242 ms
Throughput	4.1 cases/s

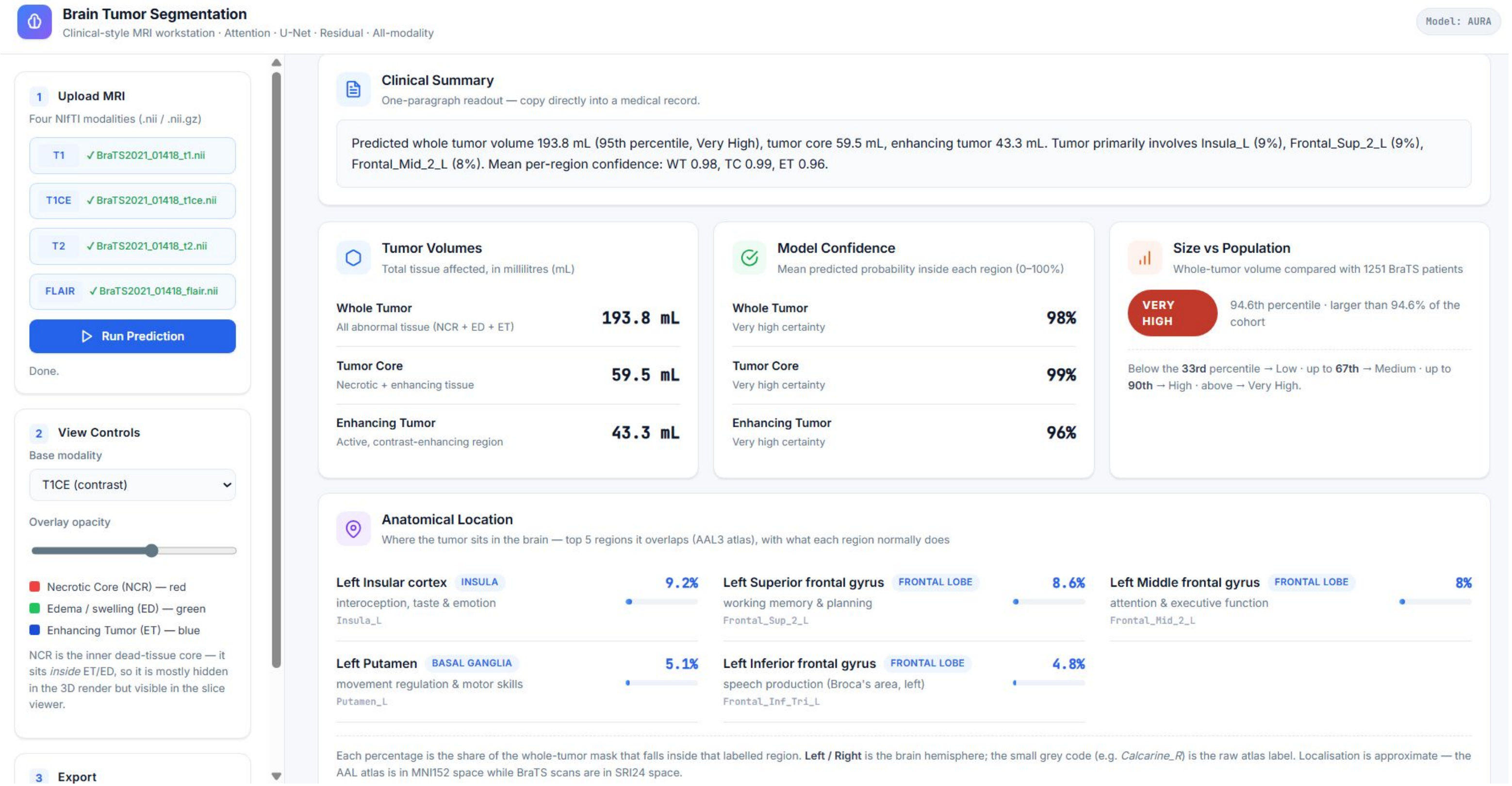
05
Deployment

The Web Workstation

- A real tool, not just a model — AURAS running live in a browser
- Capabilities: **3D view · volumes · confidence · anatomy · report**



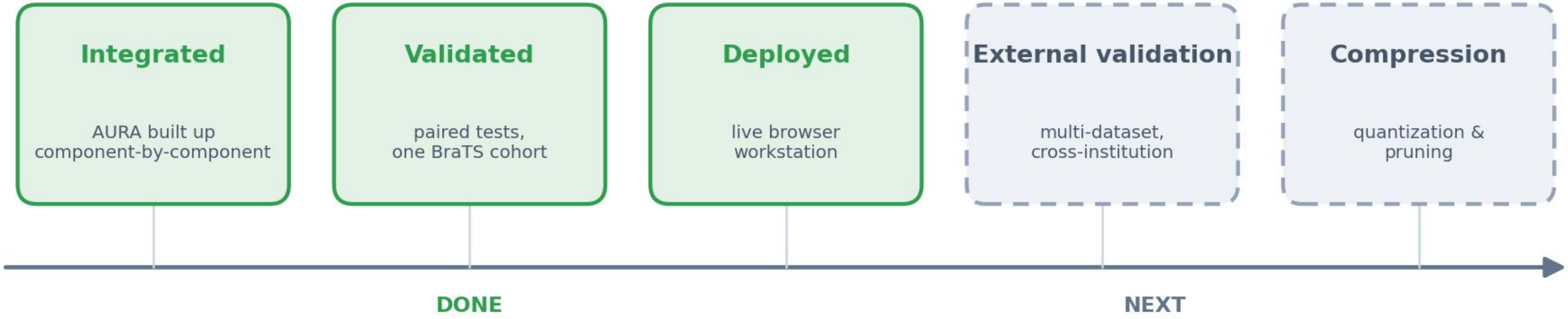
Live Demo


Brain Tumor Segmentation
Clinical-style MRI workstation · Attention · U-Net · Residual · All-modality
Model: AURA
1 Upload MRI
Four NIfTI modalities (.nii / .nii.gz)
T1 ✓BraTS2021_01418_t1.nii
T1CE ✓BraTS2021_01418_t1ce.nii
T2 ✓BraTS2021_01418_t2.nii
FLAIR ✓BraTS2021_01418_flair.nii
Run Prediction
Done.
2 View Controls
Base modality
T1CE (contrast)
Overlay opacity
Necrotic Core (NCR) — red
Edema / swelling (ED) — green
Enhancing Tumor (ET) — blue
NCR is the inner dead-tissue core — it sits inside ET/ED, so it is mostly hidden in the 3D render but visible in the slice viewer.
3 Export
Clinical Summary
One-paragraph readout — copy directly into a medical record.
Predicted whole tumor volume 193.8 mL (95th percentile, Very High), tumor core 59.5 mL, enhancing tumor 43.3 mL. Tumor primarily involves Insula_L (9%), Frontal_Sup_2_L (9%), Frontal_Mid_2_L (8%). Mean per-region confidence: WT 0.98, TC 0.99, ET 0.96.
Tumor Volumes
Total tissue affected, in millilitres (mL)
Whole Tumor
All abnormal tissue (NCR + ED + ET)
193.8 mL
Tumor Core
Necrotic + enhancing tissue
59.5 mL
Enhancing Tumor
Active, contrast-enhancing region
43.3 mL
Model Confidence
Mean predicted probability inside each region (0–100%)
Whole Tumor
Very high certainty
98%
Tumor Core
Very high certainty
99%
Enhancing Tumor
Very high certainty
96%
Size vs Population
Whole-tumor volume compared with 1251 BraTS patients
VERY HIGH
94.6th percentile · larger than 94.6% of the cohort
Below the 33rd percentile → Low · up to 67th → Medium · up to 90th → High · above → Very High.
Anatomical Location
Where the tumor sits in the brain — top 5 regions it overlaps (AAL3 atlas), with what each region normally does
Left Insular cortex INSULA 9.2%
interoception, taste & emotion
Insula_L
Left Superior frontal gyrus FRONTAL LOBE 8.6%
working memory & planning
Frontal_Sup_2_L
Left Middle frontal gyrus FRONTAL LOBE 8%
attention & executive function
Frontal_Mid_2_L
Left Putamen BASAL GANGLIA 5.1%
movement regulation & motor skills
Putamen_L
Left Inferior frontal gyrus FRONTAL LOBE 4.8%
speech production (Broca's area, left)
Frontal_Inf_Tri_L
Each percentage is the share of the whole-tumor mask that falls inside that labelled region. Left / Right is the brain hemisphere; the small grey code (e.g. Calcarine_R) is the raw atlas label. Localisation is approximate — the AAL atlas is in MNI152 space while BraTS scans are in SRI24 space.

06
Future Work

Future Work

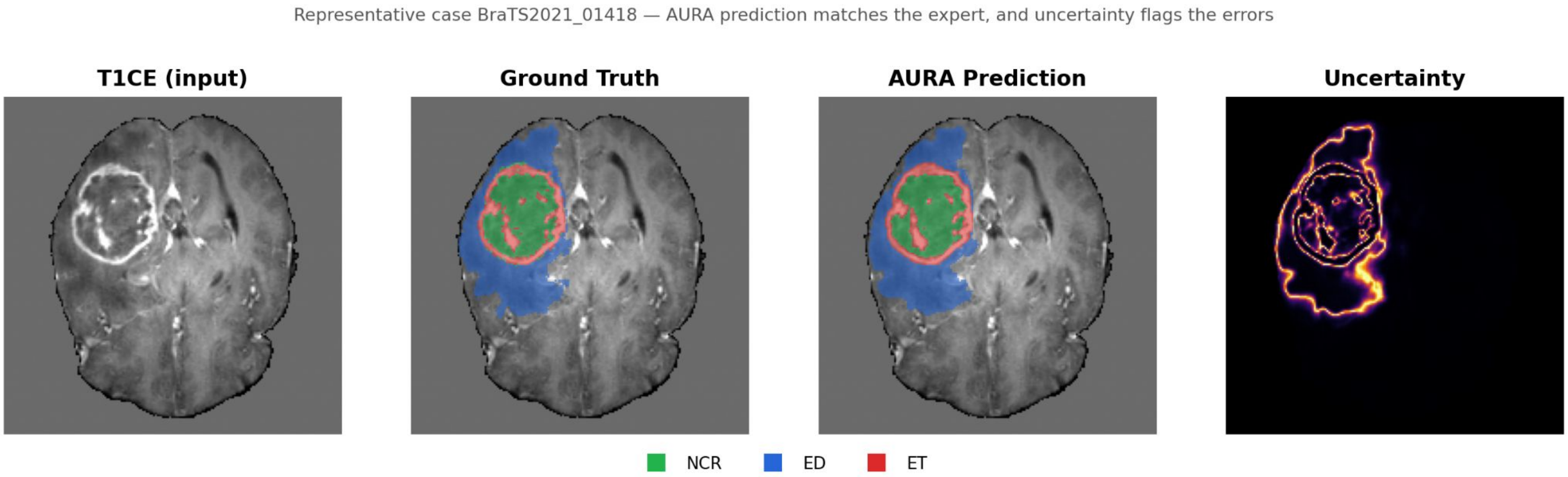
- Validate AURA across other datasets and institutions
- Quantize and prune the model for faster inference



07
Conclusion

Conclusion

- **Accurate** — strong segmentation across ET, TC, WT
- **Trustworthy** — calibrated uncertainty that tracks real error
- **Deployed** — the validated model, live in the browser



08
Q & A

Thank You for Your Time

Live Demo Video